openabel

Recover a 3D flame slice from one 2D photo, in five sessions.

PAN IIT OSS explainer • higher secondary to advanced undergraduate (levels 4-6)

Why this exists

A camera records the sum of the light along each line of sight, so a photo of a round flame hides its inside. This explainer shows the mathematics that undoes the summing, the Abel transform, and why noise makes the undoing hard.

The end goal

Recover a 3D flame slice from a 2D photo with the Abel inversion. The end goal is to recover a 3D flame slice from a 2D photo with the Abel inversion, as used for radial soot, temperature and emission profiles of laminar flames, plasma jets and sprays. The same mathematics appears in medical-style tomography of round objects and in astronomy.

How the five sessions build up

1. **Session 1: A photo adds up the light along every line of sight** (Level 1, conceptual interactive model, 60 min).
2. **Session 2: From a radial recipe to a photo: the forward Abel transform** (Level 2, mathematical engineering model, 75 min).
3. **Session 3: Back from the photo: onion peeling, noise and smoothing** (Level 3, simulation and validation, 90 min).
4. **Session 4: No peeking at the truth: choosing the smoothing from the photo alone** (Level 3, simulation and validation, 90 min).
5. **Session 5: Where is the axis? Finding the flame's symmetry axis from the photo** (Level 3, simulation and validation, 90 min).

1 Session 1: A photo adds up the light along every line of sight

A. Real-world problem

A candle flame is a 3D glowing blob, but a photo of it is flat. Each pixel does not see one point of the flame: it sees all the light made along the line from the camera through the flame. How bright is a pixel when we look at a glowing ball from the side, and why does a hollow ball look different from a solid one? (This is general education about imaging, not a measurement method for any safety decision.)

B. Concept and theory

Look at a ball of radius R from the side. A line of sight at height y above the middle passes through the ball along a straight segment called the chord. If the ball glows equally everywhere with emission eps0 (light made per centimetre of path, in arbitrary units a.u. per cm), the pixel records eps0 times the chord length. Pythagoras on the right triangle (half chord, y, R) gives the chord. For a hollow shell with empty inner radius r_in, subtract the chord through the hole.

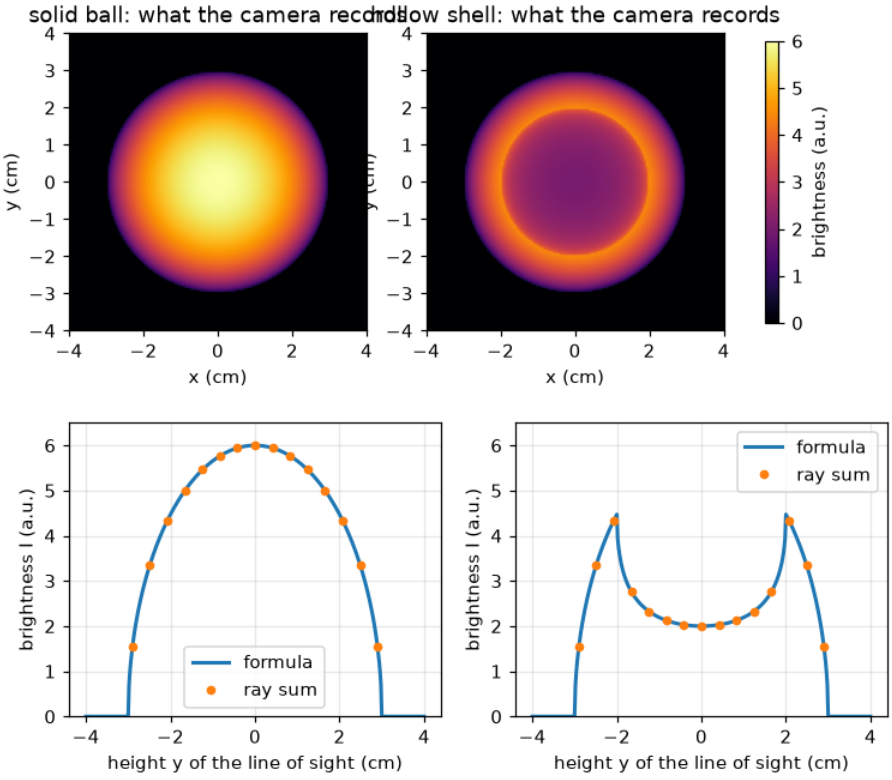
Adding up a quantity along a line is called a line integral; here emission is constant, so it is just emission times length.

- R: outer radius of the ball (cm)
- r_in: inner radius of the empty hole (0 for a solid ball) (cm)
- y: height of the line of sight above the middle (cm)
- eps0: emission: light made per cm of path (a.u. per cm)
- I(y): brightness recorded by the pixel at height y (a.u.)

$$\text{chord}(y, R) = 2\sqrt{R^2 - y^2} \quad (|y| < R), \qquad 0 \text{ otherwise}$$

$$I(y) = \varepsilon_0 \, \text{chord}(y, R) \quad \text{(solid ball)}$$

$$I(y) = \varepsilon_0 \big(\text{chord}(y, R) - \text{chord}(y, r_{\text{in}})\big) \quad \text{(hollow shell)}$$



Camera photos and brightness profiles of a solid glowing ball and a hollow shell (dots: brute-force ray sums).

E. Worked example

A solid ball of radius R = 3 cm has emission eps0 = 2 a.u. per cm. What does the camera record on a line of sight at y = 1.8 cm?

1. Square the numbers: R^2 = 9 and y^2 = 3.24.
2. R^2 - y^2 = 5.76 and its square root is 2.4 cm.

3. The chord is 2 * 2.4 = 4.8 cm.
4. Brightness = eps0 * chord = 2 a.u. per cm * 4.8 cm = 9.6 a.u.
5. Sanity check: the centre line would give 2 * 3 * 2 = 12 a.u., larger than 9.6 as it must be, and the units (a.u. per cm times cm) give a.u.

Answer. The chord is 4.8 cm and the pixel records 9.6 a.u.

F. Questions

1. **Conceptual.** Why is the middle of a solid glowing ball the brightest pixel, and why can a hollow shell be brighter at its edge than in the middle?
2. **Numerical.** A solid ball has R = 2.5 cm and eps0 = 1.2 a.u. per cm. Find the chord and the brightness at y = 1.5 cm, and at y = 0. Check the units.
3. **Graph and simulation interpretation.** In the profile of the hollow shell (R = 3 cm, r_in = 2 cm), where is the brightest point and what value does it have? Why is the centre only 2 a.u.?
4. **Design and optimisation challenge.** Design: a glowing shell has outer radius 3 cm. Choose the inner radius so that the centre brightness is at most half of the brightest brightness. Is a hole of radius 1 cm enough? Justify with a calculation.

2 Session 2: From a radial recipe to a photo: the forward Abel transform

A. Real-world problem

A real flame does not glow equally everywhere: it may glow in a bright core, or in a hollow ring around a dark centre. We describe it by an emission profile eps(r): the light made per millimetre of path at distance r from the axis. If we know eps(r), what will the camera record? And since a camera has a limited number of pixels, how many pixels across the flame do we need before our model of the photo is accurate to 1 percent?

B. Concept and theory

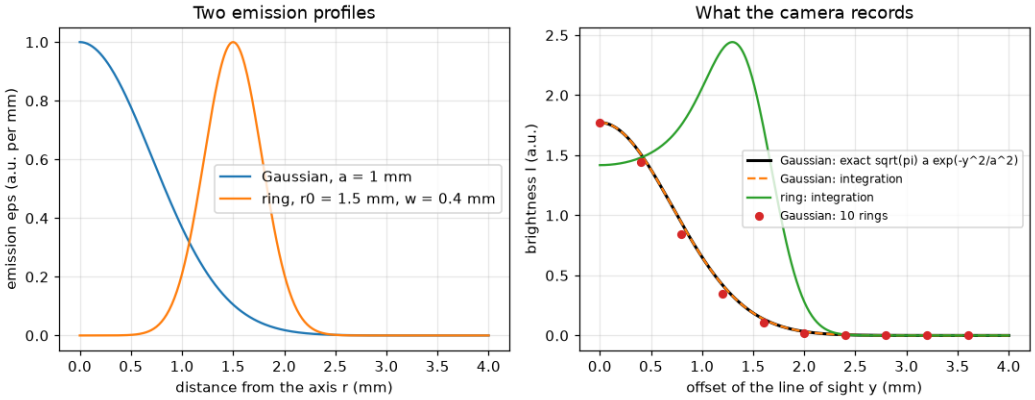
Take a line of sight at offset y from the axis. Walk along it a distance u from its closest point to the axis: there the distance from the axis is r = sqrt(y^2 + u^2). Adding the glow along the whole line (both directions) gives the brightness I(y); we keep only the region r < R, where the flame lives. This map from eps(r) to I(y) is the forward Abel transform. The integral can be written with u (no division by zero, so the trapezoid rule works). For a Gaussian eps = exp(-r^2/a^2) (a is its width) the answer is exactly a Gaussian again, I = sqrt(pi) a exp(-y^2/a^2). A camera works with pixels, so we cut the region into N rings of width dr = R/N and assume eps is constant in each ring. The line of sight at y_i = i dr crosses ring j (for j >= i) along the length W_ij; then I_i is the sum over j of W_ij eps_j, which is a matrix times a vector. Lines of sight at y_i never meet rings inside y_i, so W is zero below its diagonal.

- eps(r): emission at distance r from the axis (a.u. per mm)
- I(y): brightness recorded at offset y (a.u.)
- u: distance along the line of sight from its closest point to the axis (mm)
- R: outer radius of the modelled region (4 mm) (mm)
- a: width of the Gaussian flame (mm)
- N, dr: number of rings and ring width R/N (no unit, mm)
- W_ij: path length of line of sight i inside ring j (mm)

$$I(y) = 2\int_0^{\sqrt{R^2-y^2}} \varepsilon\Big(\sqrt{y^2+u^2}\Big)\,\mathrm{d}u = 2\int_y^R \frac{\varepsilon(r)\,r\,\mathrm{d}r}{\sqrt{r^2-y^2}}$$

$$\varepsilon(r) = e^{-r^2/a^2} \implies I(y) = \sqrt{\pi}\,a\,e^{-y^2/a^2}$$

$$W_{ij} = 2\left(\sqrt{r_{j+1}^2 - y_i^2} - \sqrt{\max(r_j^2 - y_i^2, 0)}\right), \qquad I_i = \sum_{j\geq i} W_{ij}\,\varepsilon_j$$



Left: Gaussian and ring emission profiles. Right: the photo profiles; red dots are a 10-ring model.

E. Worked example

For a Gaussian flame with a = 1 mm, what brightness does the camera record at y = 1 mm? Check it with numerical integration.

1. Use I(y) = sqrt(pi) a exp(-y^2 / a^2) with a = 1 and y = 1.
2. The exponent is -1^2 / 1^2 = -1, and exp(-1) = 0.3679.
3. sqrt(pi) = 1.7725, so I = 1.7725 * 0.3679 = 0.652 a.u.
4. Numerical check: integrate 2 exp(-(1 + u^2)) over u from 0 to 3 with the trapezoid rule; the code `abel_forward` gives 0.652.
5. Sanity check: the peak (y = 0) is sqrt(pi) = 1.77 a.u., so 0.652 is smaller, as it must be at a line of sight away from the axis.

Answer. I(1 mm) = sqrt(pi)/e = 0.652 a.u.; the numerical integral agrees.

F. Questions

1. **Conceptual.** Why must the ring matrix W be zero below its diagonal? What does that mean for solving the inverse problem?
2. **Numerical.** A Gaussian flame has width a = 0.5 mm. What is the brightness at y = 0 and at y = 0.5 mm? Give the units and a sanity check.
3. **Graph and simulation interpretation.** On the plot of ring-model error against N, how does the error change from N = 10 to N = 40 for a = 1 mm? Read the numbers and say what happens when the flame is narrower.
4. **Design and optimisation challenge.** Design: a camera images a region 8 mm wide in total (R = 4 mm on each side of the axis). The flame is a Gaussian with a = 0.7 mm and you want the ring model accurate to 1 percent of the peak. How many pixels across the whole image (both sides) and what pixel size do you need? What happens if the flame is half as wide?

3 Session 3: Back from the photo: onion peeling, noise and smoothing

A. Real-world problem

You have a photo of a flame and you want the glow of its inside, for example the radial profile of soot or temperature, not the sum along each line of sight. If the flame is round about its axis one photo is enough. But every real photo has a little noise, and the inversion makes it worse. How much worse, where, and what can you do about it without cheating? (This is a teaching model; real combustion diagnostics need calibration and many more checks.)

B. Concept and theory

Session 2 gave I = W eps, with W upper triangular. Onion peeling solves it from the outside in: the outermost ring from the outermost pixel, then each inner ring after subtracting what the outer rings already explain. With perfect data this is exact for the ring model. With noise, errors accumulate towards the axis. Because the inversion is linear, eps = Winv I, so noise of standard deviation sigma on each pixel gives noise of sigma times the length of row i of Winv on ring i; call that length the noise gain. To tame the noise we ask for a smooth answer: minimise the misfit |W eps - I|^2 plus lam times the roughness |D eps|^2, where D takes second differences (zero for a straight-line profile). Large lam gives a smooth but biased profile, lam = 0 gives noisy onion peeling. The total error combines the bias (the average error) and the noise (the scatter): total^2 = bias^2 + noise^2. To avoid an inverse crime (testing with data made by the same approximation), the test photos are made by a fine numerical integral.

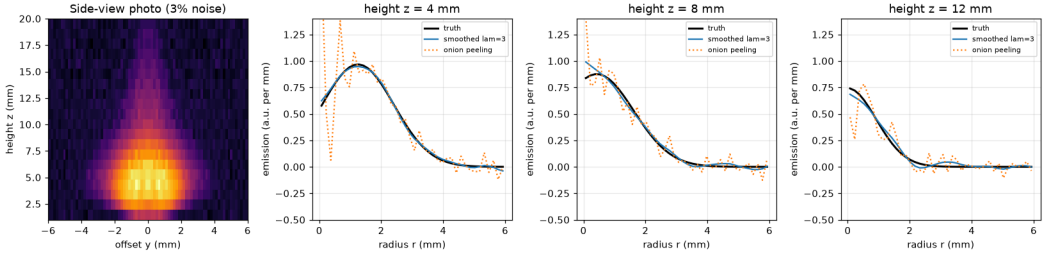
- eps_i: emission of ring i (a.u. per mm)
- I_i: brightness at offset y_i (a.u.)
- sigma: standard deviation of the photo noise (fraction of the peak brightness) (a.u.)
- gain_i: noise gain: noise in ring i divided by sigma (per mm)
- lam: smoothing strength (scaled so it has no unit) (no unit)
- bias, noise, total: RMS errors as a fraction of the peak emission (no unit)

$$I_i = \sum_{j \geq i} W_{ij} \varepsilon_j \implies \varepsilon_i = \frac{I_i - \sum_{j>i} W_{ij} \varepsilon_j}{W_{ii}}$$

$$\text{gain}_i = \Big(\sum_k (W^{-1})_{ik}^2 \Big)^{1/2}, \qquad \sigma_{\varepsilon,i} = \sigma \cdot \text{gain}_i$$

$$\hat{\varepsilon} = \arg\min_{\varepsilon} \; \|W\varepsilon - I\|^2 + \lambda s \, \|D\varepsilon\|^2 = (W^{\mathsf{T}} W + \lambda s D^{\mathsf{T}} D)^{-1} W^{\mathsf{T}} I$$

$$\text{total}^2 = \text{bias}^2 + \text{noise}^2$$



Synthetic candle photo with 3 percent noise and the recovered radial emission at three heights.

E. Worked example

Three rings of width 1 mm (R = 3 mm) give measured brightness I = (12, 9.121, 4.472) a.u. at y = 0, 1 and 2 mm. Recover the emission of each ring by onion peeling.

1. The ring matrix (Session 2 formula) has rows (2, 2, 2), (0, 3.464, 2.192) and (0, 0, 4.472).
2. Outermost ring first: eps_2 = 4.472 / 4.472 = 1.
3. Next ring: eps_1 = (9.121 - 2.192 * 1) / 3.464 = 6.929 / 3.464 = 2.
4. Innermost ring: eps_0 = (12 - 2 * 2 - 2 * 1) / 2 = 6 / 2 = 3.
5. Check by the forward model: 2*3 + 2*2 + 2*1 = 12, 3.464*2 + 2.192*1 = 9.120 and 4.472*1 = 4.472, which reproduce the measured values.

Answer. The emission is (3, 2, 1) a.u. per mm, from the axis outwards.

F. Questions

1. **Conceptual.** Why does the noise in the recovered emission grow towards the axis, and why does smoothing help but not for free?
2. **Numerical.** For N = 40 rings the noise gain of the axis ring is 5.9 per mm. The photo noise is sigma = 0.02 a.u. What is the noise in the emission of the axis ring? Compare with a Gaussian flame whose peak emission is 1 a.u. per mm.
3. **Graph and simulation interpretation.** On the trade-off graph of the Gaussian flame at 1 percent noise, how do the bias, noise and total change as lam goes from 0.00001 to 30? Where is the minimum of the total?
4. **Design and optimisation challenge.** Design: a ring-shaped flame (r0 = 1.5 mm, w = 0.4 mm) is photographed with 1 percent noise and 40 rings. Choose between lam = 0.01, 0.1 and 10 and give the total error of each (use `bias_noise_exact`). What risk does the largest lam bring for a hollow flame?

4 Session 4: No peeking at the truth: choosing the smoothing from the photo alone

A. Real-world problem

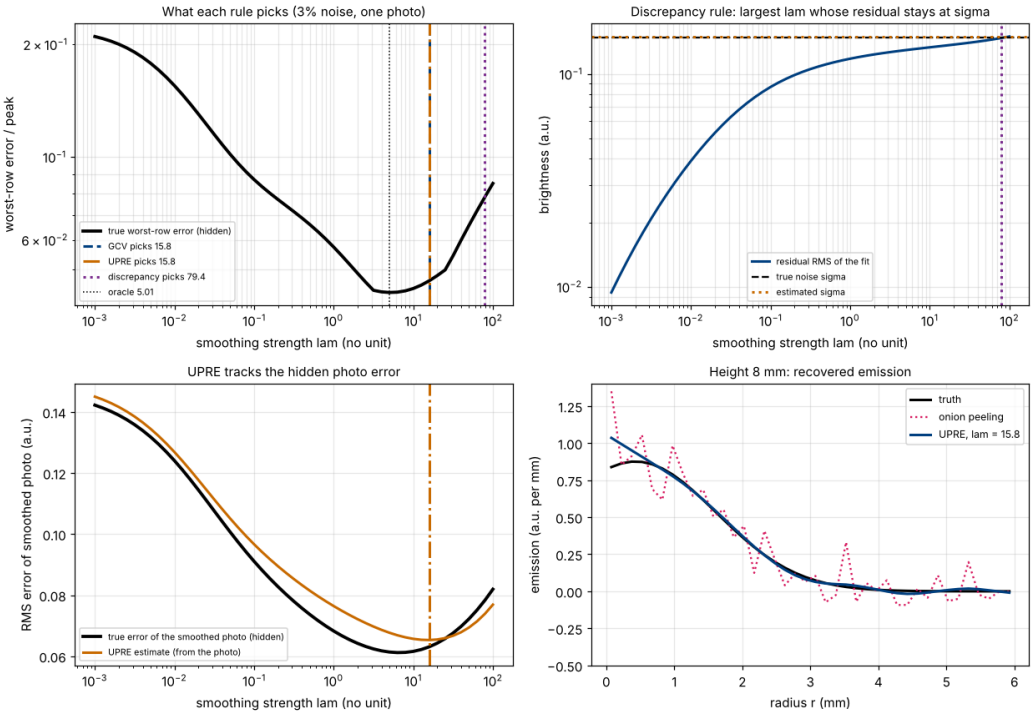
In Session 3 the smoothing strength was chosen by trying several values and keeping the one whose answer was closest to the true flame. With a real photograph nobody knows the true flame, so that recipe is impossible. What can the photo itself tell us about how much smoothing it needs? It tells us the size of its own noise, and how well a smoothed copy of it would predict a clean copy. This is general education about image processing, not a measurement method for any safety decision.

B. Concept and theory

The smoothed answer is linear in the photo, $\hat{\varepsilon} = GI$, and the smoothed photo is AI with the hat matrix $A = WG$; its trace df counts the effective parameters. A smooth profile has tiny second differences, so what remains in them is noise, and the median absolute deviation gives the noise size σ from the photo alone. Three rules then choose λ: the discrepancy principle (the largest λ with $\text{rss}/N \le \sigma^2$), UPRE (the smallest $\text{rss} + 2\sigma^2\text{df} - N\sigma^2$, an unbiased estimate of the prediction error) and generalised cross-validation (the smallest $N\,\text{rss}/(N - \text{df})^2$, which needs no σ and rests on the leave-one-out shortcut $e_i = (I_i - (AI)_i)/(1 - A_{ii})$).

$$\hat{\sigma} = \frac{1.4826}{\sqrt{6}} \operatorname{median} \left| d_i - \operatorname{median}(d) \right|, \quad d_i = I_{i+1} - 2I_i + I_{i-1}$$

$$\mathrm{UPRE}(\lambda) = \mathrm{rss} + 2\sigma^2 \,\mathrm{df} - N\sigma^2, \qquad \mathrm{GCV}(\lambda) = \frac{N \,\mathrm{rss}}{(N - \mathrm{df})^2}$$



One noisy candle photo: the error against lam and the lam each rule picks, the discrepancy rule, UPRE against the hidden error, and the recovered emission.

E. Worked example

For the height-8 row of the 3 percent candle photo (N = 40), the noise from all rows is sigma = 0.150 (sigma squared 0.0224, N sigma squared 0.896). With lam = 0.3, 3 and 15.8 the fit leaves rss = 0.596, 0.829 and 0.921 with df = 16.76, 10.11 and 7.27.

1. UPRE = rss + 2 sigma^2 df - N sigma^2 = 0.451, 0.386 and 0.351: the smallest is lam = 15.8.
2. GCV = N rss / (N - df)^2 = 0.0441, 0.0371 and 0.0344: again lam = 15.8.
3. Discrepancy: rss / N = 0.0149, 0.0207 and 0.0230 against sigma^2 = 0.0224; the largest that passes is lam = 3.
4. The true emission errors, which the rules never saw, are 0.060, 0.044 and 0.047.

Answer. UPRE and GCV prefer lam = 15.8; the discrepancy rule, limited to these three values, picks lam = 3; the true errors are 0.047 and 0.044.

F. Questions

1. **Conceptual.** Why can we not choose lam by making the residual as small as possible?
2. **Numerical.** A photo has N = 50 pixels and noise sigma = 0.2. Smoothing with lam1 leaves rss = 1.5 with df = 15; lam2 leaves rss = 1.9 with df = 9. Compute UPRE, GCV and the discrepancy test for both, and say which lam each rule prefers.

3. **Graph and simulation interpretation.** In the top right panel of the first figure, the residual of the fit reaches the dashed noise line only at lam of about 80, while the best lam is about 5. What does this tell you about the discrepancy rule?
4. **Design and optimisation challenge.** For the candle photo at 1, 3 and 6 percent noise, report the mean worst-row error of UPRE on all rows, of the oracle and of the fixed lam = 3. Which rule would you ship, and how does the chosen lam change with the noise?

5 Session 5: Where is the axis? Finding the flame's symmetry axis from the photo

A. Real-world problem

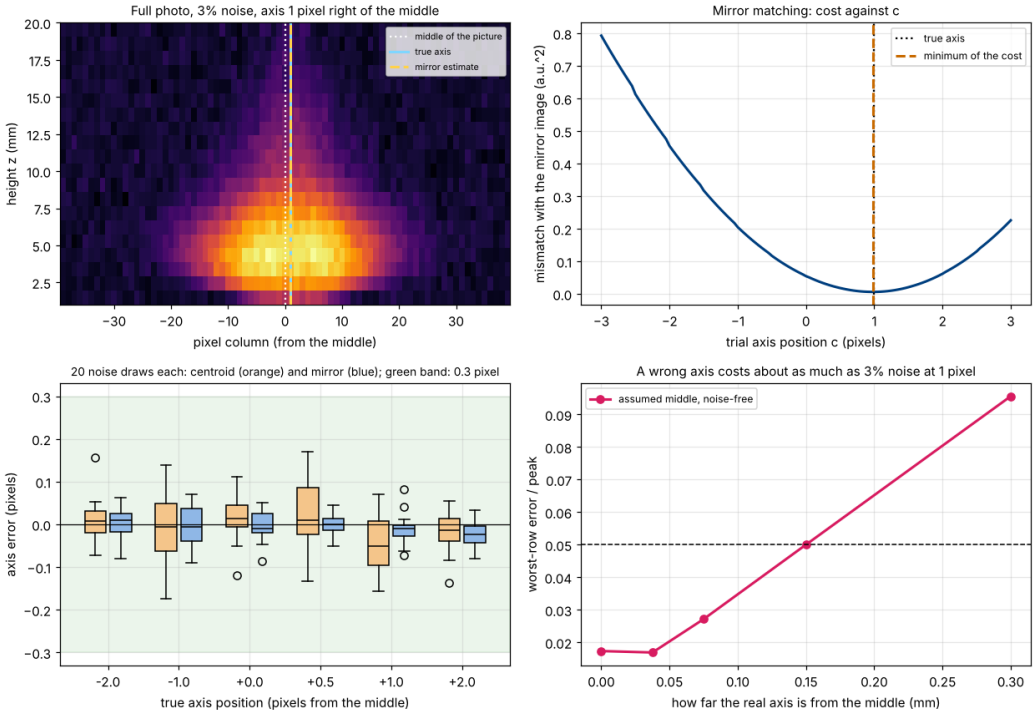
Every inversion so far assumed that the flame's axis runs exactly through the middle of one pixel column. A real camera is never aligned that well: the flame sits a fraction of a pixel to one side, and one pixel here is 0.15 mm. A one-pixel mistake already costs about as much as 3 percent noise does. Can the photo itself tell us where the axis is? Yes, because a round flame looks the same on the left and on the right of its axis. This is general education about image processing, not a measurement method for any safety decision.

B. Concept and theory

Positions are measured in pixels from the middle column; a full photo has 79 columns for 40 rings of width $dr = 0.15$ mm. If the axis is at a, a pixel at x looks along a line of sight at offset $y = |x - a|\, dr$, and its row of the ring matrix holds the chord length of each ring at that offset, so the photo never has to be resampled. Two finders read only the photo. The centroid is the centre of mass of the brightness. Mirror matching slides a trial axis c and takes the position where the photo best matches its own reflection; interpolating the reflection of a noisy photo makes the cost prefer some positions ("pixel locking"), and smoothing the photo with a Gaussian of 2 pixels first removes that. With the axis found, both halves are inverted at once, which lowers the noise of the answer.

$$a_{\text{centroid}} = \frac{\sum_x x\, I(x)}{\sum_x I(x)}, \qquad \text{cost}(c) = \operatorname*{mean}_x \big[I(x) - I(2c - x)\big]^2$$

$$W_{ij} = 2\left[\sqrt{r_{j+1}^2 - y_i^2} - \sqrt{r_j^2 - y_i^2}\right]_+, \qquad y_i = |x_i - a|\, dr$$



One noisy photo with its true and estimated axis, the mirror cost, the axis errors of both finders over 20 draws at 6 positions, and the cost of a wrong axis.

E. Worked example

One row has the values 0, 2, 6, 9, 7, 3, 1 at positions 0 to 6.

1. Centroid: sum(x I) = 90 and sum(I) = 28, so $a = 90/28 = 3.21$.
2. Mirror cost: 0.86 at $c = 3$, 4.67 at $c = 3.5$ and 14.33 at $c = 2.5$.
3. A parabola through these three points has its lowest point at $c = 3.14$.

Answer. The centroid gives 3.21, mirror matching 3.14: they agree to within 0.1 pixel (0.015 mm).

F. Questions

1. **Conceptual.** Why is the brightest pixel a poor estimate of the flame's axis?
2. **Numerical.** A row has the values 1, 4, 8, 6, 2 at positions 0 to 4. Compute the centroid, and the mirror cost at $c = 1.5$, 2 and 2.5. What does each finder say?
3. **Graph and simulation interpretation.** In the right panel of the second figure, the boxes "mirror, one half" and "mirror, both halves" use the same axis, yet the first is higher. What does this tell you?
4. **Design and optimisation challenge.** For a candle photo with 3 percent noise and the axis 0.5, 1 and 2 pixels from the middle, report the mean worst-row error of "assume the middle (one half)", of "mirror axis, both halves" and of "true axis, both halves". When can you skip the axis search?

6 Real-world mini-project

Recover the inside of a candle-like flame from a synthetic photo. A lab gives you a synthetic side-view photograph of a candle-like flame with 3 percent noise (`candle_photo(N=40,`

`frac=0.03)` in Session 3). The flame is hollow near its base and fills in higher up. Recover the radial emission at three heights (z = 4, 8 and 12 mm) with `invert_row`, trying the smoothing strengths in `LAM_GRID`, and choose the one with the smallest worst-row error. Because this is a synthetic photo, the true emission `candle_emission` is known, so you can score each choice.

Choose the smoothing for the candle without peeking (classroom exercise only). Given only the noisy 3 percent candle photo, pick one smoothing strength for all rows without using the true emission, then grade the rule on 20 noise draws against the best possible strength (the oracle) and the fixed lam = 3 of Session 3. With UPRE on all rows the mean worst-row error is 0.049 (oracle 0.044, fixed lam = 3 0.053); GCV applied to one row at a time is erratic (mean 0.066, worst 0.175); the line "0.05 on every draw" cannot be met by any rule, since even the oracle reaches it on only 14 of 20 draws. This is a teaching exercise on synthetic data, not a method for measuring a real flame.

Find the axis of the candle photo and invert with it (classroom exercise only). Given a noisy 3 percent photo of the candle whose axis is 1 pixel (0.15 mm) off the middle, find the axis from the photo alone and invert with both halves, then grade the pipeline on 20 noise draws. With the axis found by mirror matching the mean worst-row error is 0.041 (the same as with the true axis), against 0.065 when the middle is assumed; the centroid gives 0.041 too but scatters more. Using one half only costs 0.053 even with the axis found. The line "at or below 0.045" is met on average, not on every draw: 13 of 20 draws, also with the true axis. This is a teaching exercise on synthetic data, not a method for measuring a real flame.

7 The path ahead

1. **Session 6: Use neighbouring rows.** A whole-image inversion of all 20 rows with less noise than row-by-row inversion.
2. **Session 7: Keep the hollow base hollow.** An inversion that keeps the dark core of the candle base without negative values.
3. **Session 8: Render it in 3D and animate.** A looping 3D cut-away animation of the recovered flame.

8 Answers and hints

Session 1

1. **Conceptual.** Hint: Compare the path lengths through glowing material for a line through the middle and a line near the edge of the hole. Solution: The pixel adds up light along the path, so a longer path through glowing material gives a brighter pixel. In a solid ball the line through the middle is the longest. In a shell the middle line spends most of its length in the empty hole, so only the short walls count; a line that just grazes the hole travels a long way inside the wall, so the brightest line is the one that touches the inner radius.
2. **Numerical.** Hint: Use chord = 2 sqrt(R^2 - y^2): at y = 1.5, R^2 - y^2 = 6.25 - 2.25 = 4. Solution: At y = 1.5 cm: chord = 2 * sqrt(4) = 4 cm and I = 1.2 * 4 = 4.8 a.u. At y = 0: chord = 2R = 5 cm and I = 1.2 * 5 = 6.0 a.u. Units: (a.u. per cm) times cm = a.u. Sanity: the off-centre line is shorter so it is dimmer.
3. **Graph and simulation interpretation.** Hint: The brightest line touches the inner circle: y = r_in. Evaluate 2 sqrt(R^2 - r_in^2). Solution: The brightest point is at y = 2 cm with I = 2 sqrt(9 - 4) = 2 sqrt(5) = 4.47 a.u. The centre line has chord 6 cm through the outer ball minus 4 cm through the hole, so 2 cm and I = 2 a.u. The profile therefore has a dip in the middle and two bright shoulders, called limb brightening.

4. **Design and optimisation challenge.** Hint: The ratio I(0) / I_max = sqrt((R - r_in) / (R + r_in)). Set it to 1/2. Solution: I(0) / I_max = sqrt((R - r_in) / (R + r_in)) <= 1/2 means (R - r_in) / (R + r_in) <= 1/4, so r_in >= 0.6 R = 1.8 cm. A hole of 1 cm is not enough: the ratio is sqrt(2/4) = 0.71. With r_in = 2 cm the ratio is sqrt(1/5) = 0.45, which works. Trade-off: a bigger hole gives a stronger dark centre but a thinner glowing wall, so the whole image is dimmer.

Session 2

1. **Conceptual.** Hint: Which rings can the line of sight at y_i cross? Solution: A line of sight at offset y_i stays at a distance of at least y_i from the axis, so it can only cross rings whose outer edge is beyond y_i, that is rings j >= i. Rings inside y_i give W_ij = 0. Because W is triangular, the outermost line of sight meets only the outermost ring, the next one meets two rings, and so on: the inverse can be solved ring by ring from the outside inwards (onion peeling, Session 3).
2. **Numerical.** Hint: I(y) = sqrt(pi) a exp(-y^2/a^2). Solution: At y = 0: I = 1.7725 * 0.5 = 0.886 a.u. At y = 0.5 mm: I = 0.886 * exp(-1) = 0.326 a.u. Units: eps in a.u. per mm times mm gives a.u. Sanity: a narrower flame has a smaller total (0.886 against 1.77 for a = 1 mm) and the value falls to 1/e = 37 percent of the peak at y = a.
3. **Graph and simulation interpretation.** Hint: The checkpoint prints the exact values; the curves for a = 0.5 and a = 0.7 are above the one for a = 1. Solution: For a = 1 mm the error is 5.4 percent at N = 10, 2.0 percent at N = 20 and 0.7 percent at N = 40: it falls roughly like dr to the power 1.4 (each halving of the pixel size divides the error by about 2.7). Narrower flames have larger errors at the same N because the profile changes faster across a ring; the a = 0.5 mm curve needs about twice as many rings to reach the same error.
4. **Design and optimisation challenge.** Hint: Use `smallest_rings(0.7)`; pixels across = 2N; pixel size = R / N. Solution: smallest_rings(0.7) = 47 rings, so the 8 mm wide image needs 2 * 47 = 94 pixels across, with pixel size 4 / 47 = 0.085 mm. For a = 0.5 mm the answer is 65 rings (pixel 0.062 mm); for a = 1 mm it is 33 rings (pixel 0.121 mm). In all three cases the pixel size is about 0.12 times the width a, that is about 8 pixels across one Gaussian width a. Trade-off: finer pixels need a closer or better camera and gather less light per pixel, which raises the noise (Session 3).

Session 3

1. **Conceptual.** Hint: Think about how many pixels' errors are added up in the innermost ring. Solution: Each inner ring is found by subtracting the contribution of all outer rings from the pixel, so the errors of all the outer solutions pile up, and then they are divided by W_ii = 2 dr sqrt(2i + 1), which is smallest for the axis ring (i = 0). The noise gain is about 5.9 per mm at the axis and 0.56 per mm at the edge (N = 40). Smoothing averages neighbouring rings, which cancels random noise, but it also averages real structure, so sharp features (a peak, a hole) are lowered and blurred: this is the bias. The total error is smallest when the two are balanced.
2. **Numerical.** Hint: noise = sigma * gain; then divide by the peak. Solution: noise = 0.02 a.u. * 5.9 per mm = 0.12 a.u. per mm. Relative to the peak of 1 a.u. per mm that is 12 percent, six times the 2 percent noise of the photo. The outer ring has gain 0.56 per mm and noise 0.011 a.u. per mm (1 percent). Units: a.u. times per mm is a.u. per mm, the unit of the emission.
3. **Graph and simulation interpretation.** Hint: Move the log10 lam slider in the app or read the third panel of the figure. Solution: The bias is flat at about 0.4 percent for small lam and rises to about 1.9 percent at lam = 30. The noise falls steadily from about 2.7 percent to about 0.4 percent. The total is about 2.7 percent for tiny lam, falls to a minimum of about

1.1 percent near lam = 0.3 (log10 lam = -0.5), then rises again as the bias takes over. The best choice is the one at the valley, not the largest lam.

4. **Design and optimisation challenge.** Hint: Print `bias_noise_exact(lambda q: ring_eps(q, 1.5, 0.4), 40, lam, 0.01)` for the three values. Solution: The total errors are about 2.7 percent for lam = 0.01 (noise dominates), 2.0 percent for lam = 0.1 (best) and 7.7 percent for lam = 10 (bias dominates). Choose lam = 0.1. With lam = 10 or more, smoothing lowers the ring peak and fills or even drives negative the dark hole on the axis, so the flame would be misread as less hollow and with negative emission, which is unphysical. Trade-off: a ring flame needs less smoothing than a Gaussian flame because it has sharper structure.

Session 4

1. **Conceptual.** Hint: What lam gives the smallest possible residual, and what does the recovered profile look like then? Solution: The residual is zero at lam = 0, where the smoothed photo is the photo itself and the inversion is plain onion peeling: it fits the noise perfectly and amplifies it into a wild profile. A good choice leaves a residual about as large as the noise. UPRE and GCV correct the residual for the df numbers the smoother spends on fitting noise.
2. **Numerical.** Hint: UPRE = rss + 2 sigma^2 df - N sigma^2 with sigma^2 = 0.04 and N sigma^2 = 2.0. Solution: lam1: UPRE = 1.5 + 1.2 - 2.0 = 0.70, GCV = 50 x 1.5 / 35^2 = 0.0612, rss / N = 0.030 passes. lam2: UPRE = 1.9 + 0.72 - 2.0 = 0.62, GCV = 50 x 1.9 / 41^2 = 0.0565, rss / N = 0.038 passes. UPRE and GCV prefer lam2, and the discrepancy rule takes the largest lam that passes, also lam2.
3. **Graph and simulation interpretation.** Hint: How large is the residual of a perfect smoother compared with sigma? Solution: A smoother that uses df numbers fits part of the noise, so even a perfect one leaves a mean-square residual of about sigma^2 (N - df) / N, roughly 0.78 sigma^2 at the best lam. The discrepancy rule waits until the residual reaches sigma^2, which needs a far stronger lam, so it over-smooths.
4. **Design and optimisation challenge.** Hint: Also print the median of choose_lam(img, "upre") over the draws for each noise level. Solution: Mean worst-row error of UPRE, oracle and fixed lam = 3: 0.023, 0.021, 0.024 at 1 percent; 0.049, 0.044, 0.053 at 3 percent; 0.080, 0.072, 0.102 at 6 percent. The median lam chosen by UPRE grows with the noise (about 1.6, 10 and 40). Ship UPRE on all rows (or GCV if sigma cannot be trusted): it is within about 10 percent of the oracle and adapts, while the fixed value is tuned for one noise level.

Session 5

1. **Conceptual.** Hint: Look at the profile of the flame at 4 mm height, and think about what one noisy pixel is worth. Solution: The flame is a hollow glow ring at its base, so at 4 mm the brightest pixel is 4 pixels (0.6 mm) off the axis, and low in the flame the profile is flat on top, so one noisy pixel moves the maximum by pixels. The centroid and the mirror cost use every pixel and every row.
2. **Numerical.** Hint: The mirror of x about c is 2c - x. Solution: Centroid 46/21 = 2.19. Cost 2 at c = 2, 4 at c = 2.5 and 20.5 at c = 1.5; a parabola through them has its lowest point at 2.20. Both finders say about 2.2 pixels.
3. **Graph and simulation interpretation.** Hint: How many independent noisy pixels does each inversion use? Solution: One half uses about 40 pixels and both halves 79, with independent noise, so the noise of the answer falls by about the square root of 2 (mean error 0.053 to 0.041). Finding the axis is what makes the second half usable.
4. **Design and optimisation challenge.** Hint: Compare the cost of a wrong axis with the noise floor, about 0.041. Solution: Assuming the middle gives 0.048, 0.065 and 0.110 for 0.5, 1 and 2 pixels; mirror axis with both halves 0.041, 0.041 and 0.044; true axis with both halves 0.042, 0.041 and 0.043. The search costs nothing measurable. Skip it only if the camera is

aligned within about 0.25 pixel, and then you also give up the second half.

9 How to join

Read `CONTRIBUTING.md` in the repository, then pick an issue labelled `good first issue`. Repository: https://github.com/KrunalPanchal1995/pan-iit-oss/tree/main/openabel